Lecture 19

Particle in a Box

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Last time, we set out to solve the Schrödinger equation

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} + V(x)\psi(x,t) \tag{1}$$

Tried Separable Solutions

$$\psi(x,t) = \phi(x)\,T(t)$$

Then we saw that

$$T(t) = e^{-iEt/\hbar}$$

and

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)\right]\phi(x) = E\phi(x) \tag{2}$$

Some terminology: Eq (1) above is called time-dependent Schrodinger Equation and Eq(2) above is called time-independent Schrodinger equation.

Why are these special separable solutions useful?

This is a short digression to tell you why separable solutions are useful. While solving (2) we will see that many solutions exist, one for each value of E. If we use n to label various possible values of E: E_n, then we get a $\phi_n(x)$ for each E_n. Then each separable solution can be written as

$$\psi_n(x,t) = \phi_n(x)e^{-iE_nt/\hbar}$$

Turns out any general solution to (1) can be written as a linear combination of $\phi_n(x)e^{-iE_nt/\hbar}$. That is the general solution $\psi(x,t)$ can be expressed as

$$\boxed{\psi(x,t) = \sum_n C_n\phi_n(x)e^{-iE_nt/\hbar}}$$

Any wavefunction $\psi(x,t)$ can be expressed in the above form. It is just a matter of finding the constants C_n. Lets leave this digression here and get back to finding separable solutions.

So far, we have the time-independent Schrödinger equation

$$-\frac{\hbar^2}{2m}\frac{d^2\phi}{dx^2} + V(x)\phi = E\phi.$$

For $V(x) = 0$, this equation indeed gives plane waves $e^{\pm ikx}$ which we have already encountered while discussing the momentum operator. Those correspond to free particles with definite momentum and energy $E = \hbar^2k^2/2m$.

But now, let us look at something more interesting, cases where the particle is confined by a potential.

In such systems, the allowed energies are no longer continuous but quantized.

The simplest and most instructive example of confinement is a particle trapped in an infinite potential well: the one-dimensional particle in a box.

PARTICLE IN A ONE-DIMENSIONAL BOX: SETUP AND BOUNDARY CONDITIONS

A central task in quantum mechanics is to use the time-independent Schrödinger equation to determine the allowed energies and the corresponding stationary-state wavefunctions for a given potential $V(x)$. In this section we study the simplest example of a bound system: a particle confined to a finite interval of space so that it cannot escape to infinity.

The model

Consider a particle constrained to move only along the x-axis, between two perfectly rigid walls located at $x = 0$ and $x = L$. The potential energy describing this situation is

$$V(x) = \begin{cases} 0, & 0 < x < L, \\ \infty, & x \leq 0 \text{ or } x \geq L. \end{cases}$$

Because the walls are infinitely high, the particle cannot penetrate or escape through them. Between the walls, the particle is free ($V = 0$). This idealized system is often referred to as a particle in a box, and it serves as a simple model for an electron confined inside a long molecule, a narrow wire, or a quantum well.

Wavefunction requirements

Let $\psi(x)$ denote a stationary-state wavefunction. Since the particle cannot be found outside the region $0 \leq x \leq L$, we must have

$$\psi(x) = 0 \qquad \text{for } x \leq 0 \text{ and } x \geq L.$$

This requirement is also enforced by the Schrödinger equation itself: the term $V(x)\psi(x)$ would be infinite unless ψ vanishes wherever $V(x) = \infty$.

In addition:

1. $\psi(x)$ must be continuous on the entire real line. A discontinuous wavefunction cannot satisfy the Schrödinger equation.
2. At the boundaries, continuity of ψ requires

$$\psi(0) = 0, \qquad \psi(L) = 0.$$

These are the boundary conditions that determine the allowed stationary states.

Schrödinger equation inside the well

Inside the region $0 < x < L$ the potential vanishes, so the time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi(x).$$

This is the same differential equation as for a free particle, so at first sight one might imagine that plane waves such as $\psi(x) = Ae^{ikx}$ would be acceptable solutions. Indeed, Ae^{ikx} is continuous, differentiable, and satisfies the equation with

$$E = \frac{\hbar^2 k^2}{2m}.$$

However, such a plane wave fails completely to satisfy the boundary conditions:

$$\psi(0) = A \neq 0, \qquad \psi(L) = Ae^{ikL} \neq 0.$$

The only way to force these to vanish is to set $A = 0$, which would eliminate the particle entirely. Thus a free-particle plane wave cannot represent a particle confined between rigid walls.

Sine-wave solutions

We therefore look for solutions that satisfy both the differential equation and the boundary conditions. The general solution of the inside equation is

$$\psi(x) = A\sin(kx) + B\cos(kx), \qquad k = \sqrt{\frac{2mE}{\hbar^2}}.$$

The condition $\psi(0) = 0$ forces $B = 0$, leaving

$$\psi(x) = A\sin(kx).$$

To satisfy $\psi(L) = 0$ we must have

$$\sin(kL) = 0 \quad \Longrightarrow \quad kL = n\pi, \qquad n = 1, 2, 3, \ldots$$

so the allowed values of k are

$$k_n = \frac{n\pi}{L}.$$

The corresponding wavelengths

$$\lambda_n = \frac{2\pi}{k_n} = \frac{2L}{n},$$

show that the length of the well contains an integral number of half-wavelengths, exactly as in the normal modes of a stretched string fixed at both ends.

Thus the stationary-state wavefunctions inside the well are

$$\psi_n(x) = A\sin\left(\frac{n\pi x}{L}\right), \qquad n = 1, 2, 3, \ldots$$

and $\psi_n(x) = 0$ outside the interval.

Energy levels for a particle in a box

Inside the well the particle is free, so its total energy is purely kinetic:

$$E = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m},$$

where

$$p = \hbar k = \frac{h}{\lambda}$$

is the magnitude of the momentum associated with a plane wave of wave number k and wavelength λ. This is exactly the momentum–energy relation for a free particle, which is appropriate because $U(x) = 0$ in the region $0 < x < L$.

For the allowed stationary states of the box we have already found that

$$k_n = \frac{n\pi}{L}, \qquad n = 1, 2, 3, \ldots$$

corresponding to wavelengths

$$\lambda_n = \frac{2\pi}{k_n} = \frac{2L}{n}.$$

Hence the allowed momentum magnitudes are

$$p_n = \hbar k_n = \frac{n\pi\hbar}{L} = \frac{nh}{2L}.$$

Substituting p_n into the kinetic-energy formula gives the quantized energy levels

$$E_n = \frac{p_n^2}{2m} = \frac{n^2 h^2}{8mL^2} = \frac{n^2\pi^2\hbar^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots$$

Thus each allowed energy level is labeled by the integer n, and higher values of n give higher energies that grow as n^2.

At first glance it might seem possible for the particle to have $E = 0$. From the expression

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2},$$

this would formally correspond to choosing $n = 0$.

However, substituting $n = 0$ into the spatial wavefunction

$$\phi_n(x) = C \sin\left(\frac{n\pi x}{L}\right)$$

yields the identically zero function. A wavefunction that is zero everywhere does not represent a physical state. Hence $n = 0$ is not an allowed quantum number, and the energy can never vanish.

This result has a clear physical interpretation. A state with $E = 0$ would imply $p = 0$ and therefore a definite momentum. According to the Heisenberg uncertainty principle, a particle with precisely defined momentum would have an infinite uncertainty in position and could be found anywhere along the x-axis. But in the present system the particle is confined to the finite interval $0 \le x \le L$;

its position is restricted, so its momentum cannot be exactly zero. Thus the lowest allowed energy is

$$E_1 = \frac{\pi^2 \hbar^2}{2mL^2},$$

a purely quantum "zero-point" energy arising from confinement.

It is also worth noting that none of the allowed stationary-state wavefunctions represents a state of definite momentum. Each ϕ_n is a real function and may be written as an equal superposition of right-moving and left-moving plane waves with momenta $\pm p_n$:

$$p_n = \frac{nh}{2L}.$$

This superposition has a definite energy but no definite momentum, and is the reason why the particle can remain confined within the box.

Corresponding stationary-state wavefunctions

For each quantum number n the eigenfunction is obtained by substituting $k_n = n\pi/L$ into the earlier solution $\sin(kx)$:

$$\psi_n(x) = C \sin\left(\frac{n\pi x}{L}\right), \qquad n = 1, 2, 3, \ldots$$

where C is the normalization constant (to be determined shortly). Each value of n therefore corresponds to a specific momentum magnitude, wavelength, energy, and stationary-state wavefunction.

Normalization and restoring the time dependence

Up to this point we have determined the spatial part of the stationary-state wavefunctions inside the well:

$$\psi_n(x) = C \sin\left(\frac{n\pi x}{L}\right), \qquad n = 1, 2, 3, \ldots$$

with $\psi_n(x) = 0$ outside the interval $[0, L]$. The constant C is fixed by the requirement that the wavefunction represent a valid probability amplitude.

Normalization

A physically acceptable wavefunction must satisfy

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = 1.$$

Since ψ_n vanishes outside the box, this becomes

$$\int_0^L |\psi_n(x)|^2 \, dx = 1.$$

Substituting the expression for ψ_n gives

$$|C|^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = 1.$$

The integral is straightforward:

$$\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2}.$$

Hence

$$|C|^2 \frac{L}{2} = 1, \qquad \Longrightarrow \qquad C = \sqrt{\frac{2}{L}}.$$

Thus the normalized stationary-state wavefunctions are

$$\boxed{\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \qquad n = 1, 2, 3, \ldots}$$

Restoring the time dependence

The full time-dependent Schrödinger equation separates as

$$\psi_n(x,t) = \phi_n(x) e^{-iE_n t/\hbar},$$

where E_n is the corresponding energy eigenvalue,

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}.$$

Thus the complete stationary-state solutions for the particle in a box are

$$\boxed{\psi_n(x,t) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) e^{-iE_n t/\hbar}, \qquad n = 1, 2, 3, \ldots}$$

Each such state has a probability density

$$|\psi_n(x,t)|^2 = |\phi_n(x)|^2,$$

which is independent of time. This is why solutions of this form are called stationary states: although the wavefunction oscillates in time through the phase factor $e^{-iE_n t/\hbar}$, all observable probabilities remain constant.